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A Additional Analyses of Results from Li et al. (2021)

As mentioned in Footnote 3, Li et al. (2021) conducted two more experiments that *prima facie* provided additional evidence for implicit meaning representations and state tracking abilities in language models. However, the data and setup of these two experiments also likely overestimates models' abilities.

In the second set of probing classifier experiments, Li et al. (2021) used data generated using the TextWorld engine (Côté et al., 2019). In this setup, there is a textual description of several entities in a text-based game (e.g., *a wooden door*) as well as actions that a player took (e.g., *opening the wooden door*). Their probing classifier takes the representations of either one entity and a property of that entity (e.g., that *the wooden door is closed*) or the representations of two entities and a relation between them (e.g., that the *king-sized bed is in the bedroom*) and from these representations, the classifier has to predict whether a given proposition is true or false considering the initial description and the series of actions that a player took. Only propositions that involve entities that have been mentioned are probed. The issue with this setup is that there are many propositions that are always true in both the training and evaluation splits (e.g., in all the game simulations, the chest drawer is in the bedroom, so the probing classifier should always return true for this input independent of the previous context). Furthermore, even for the entity-property propositions and entity-relation-entity propositions which are not always true in the training and evaluation data, the data is very biased and a baseline that predicts the most common answer in the training data without taking the initial descriptions or the user actions into account (a violation of Desideratum 3, see Section 3.1), already achieves an accuracy of 88.5%, a number that puts the reported probing classifier accuracy of 96.9% into a bit more context.

Further, Li et al. (2021), also presented an experiment where they manipulated specific entity representations of a synthetic version of the Alchemy dataset. In this experiment, they first encoded an initial description D and an operation O which affected a beaker X using a language model that had been finetuned to predict the next operation, resulting in representation R_1 . Then, they encoded the same initial representation D and an operation of

the form *Drain n from beaker Y* , where beaker Y was always different from beaker X and n was the amount of liquid that was in beaker Y according to the initial description. This resulted in representation R_2 . Then, they extracted the representation of the initial description of beaker Y from representation R_2 and replaced the representation of beaker Y in R_1 with the corresponding representation in R_2 to obtain R_{mixed} . They then used R_{mixed} as an input to T5 and showed that the predicted next operation based on R_{mixed} was considerably more often compatible with both operations (the operation encoded in R_1 and the operation encoded in R_2) compared to predicting the next operation from R_1 or R_2 , which they took as evidence that the token representations of the initial description encoded the entities state after performing the operation. The issue with this experiment is that the operation from which R_2 was computed was always of the form *Drain n from Y th beaker*, so the final state of beaker Y was always empty. Therefore, this experiment primarily shows that the token *drain* affected the representation of the initial state (and subsequently the prediction of the next operation) but not more generally, that the actual state of the manipulated beaker Y is fully encoded in its initial state description. To answer that question, one would have to repeat this experiment with more complex operations that do not give away the final state (a violation of Desideratum 2, see Section 3.1).

In summary, the additional experiments in Li et al. (2021) also violate some of the desiderata we laid out in Section 3.1, and therefore it is difficult to draw conclusions about state tracking abilities from these experiments.

B Example Input-Output Pairs

(3) shows an example input-output pair from our base dataset (NumOps on Box 6 = 2).

- (3) a. INPUT: *Box 0 contains the painting, Box 1 contains the bell, Box 2 contains the guitar, Box 3 contains the egg and the mirror and the sheet, Box 4 contains the chemical, Box 5 contains the disk and the wire, Box 6 contains the glass and the knife. Move the glass from Box 6 to Box 4. Put the gift into Box 5. Move the guitar from Box 2 to Box 6. Put the milk into Box 4. Remove the mirror and the sheet from*

Box 3. Box 6 __

- b. OUTPUT: *contains the guitar and the knife.*

(4) shows an example input-output pair from the dataset with operations that contain referring expressions that are ambiguous without considering the initial state and the previous operations (NumOps on Box 6 = 2). Note that in order to correctly interpret *Move the guitar from Box 2 to Box 6*, the model has to consider the information that the blue guitar (as opposed to the red guitar) was in Box 2 prior to the operation. Hence, this operation is ambiguous out of context.

- (4) a. INPUT: *Box 0 contains the yellow book and the green flower and the red guitar, Box 1 contains the small bomb and the small book and the blue bone, Box 2 contains the blue guitar, Box 3 contains the blue bell, Box 4 contains the green paper and the yellow note and the yellow television, Box 5 contains the yellow bell, Box 6 is empty. Move the guitar from Box 2 to Box 6. Put the blue wire and the big television into Box 5. Move the flower from Box 0 to Box 6. Box 6* __
- b. OUTPUT: *contains the blue guitar and the green flower.*

(5) shows an example input-output pair from the dataset with the *Move contents of* operation (NumOps on Box 6 = 2). Note that in order to correctly interpret *Move the contents of Box 2 to Box 6*, the model has to consider the information that the tea (as opposed to the red guitar) was in Box 2 prior to the operation.

- (5) a. INPUT: *Box 0 contains the fan and the gift and the letter, Box 1 contains the beer and the mirror and the tie, Box 2 contains the tea, Box 3 contains the boot, Box 4 contains the coat and the plate and the shirt, Box 5 contains the bottle, Box 6 is empty. Move the contents of Box 2 to Box 6. Put the dress and the painting into Box 5. Move the letter from Box 0 to Box 6. Box 6* __
- b. OUTPUT: *contains the letter and the tea.*